Day 4:- GPUs and Cooling Systems The Theory : Basics of GPUs and Cooling Systems : Practical : Identify a GPU and Cooling System in a PC setup.

Ans. :-

Tasks 1 :- Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

1. **GPU 1 (Integrated Graphics) :**
    - **Brand :-** Intel
    - **Model :-** Intel(R) HD Graphics Family
    - **Type :-** Integrated GPU

2. **GPU 2 (Dedicated Graphics) :**
    - **Brand :-** NVIDIA
    - **Model :-** NVIDIA GeForce GT 720M
    - **Type :-** Dedicated GPU

Tasks 2 :- Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

Radiator + Fans (TOP Exhaust) :- The radiator dissipates heat from the liquid coolant , while the fans push hot air out of case.

Rear Exhaust Fan :- Expels hot air from inside the case to maintain proper airflow and reduce overall temperature.

CPU Water Block /Pump :- Absorbs heat from the CPU and circulates coolant through the loop to the radiator.

Coolant Tubes :- Carry the liquid coolant between the CPU block and the radiator.

GPU Heatsink & Fans :- The graphics card cooler dissipates heat from the GPU using fans and heatsink.

Front Intake Fans :- Draw Draw cool air into the case to provide fresh airflow for all components.

Tasks 3 :- Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling	Liquid Cooling
Cost	Less expensive and budget-friendly.	More expensive due to pumps, radiators, and tubing.
Cooling Efficiency	Good for most users but less effective under heavy loads.	Good for most users but less effective under heavy loads.
Maintenance	Minimal maintenance; mainly dust cleaning.	Minimal maintenance; mainly dust cleaning.
Installation	Easy to install and replace.	Easy to install and replace.

Tasks 4 :- Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

I would choose liquid cooling for the GPU. Liquid cooling is more effective at removing heat, which helps maintain lower GPU temperatures during long gaming and IPL streaming sessions.

It also tends to operate more quietly than air cooling, making the streaming experience better. Although liquid cooling is more expensive and requires more maintenance, its superior cooling performance makes it a good choice for a high-performance gaming and streaming PC.